APPLICATION DEVELOPMENT WORK SAMPLES & TECHNICAL DOCUMENTATION

Candidate: Bryan John C. Lopez
Role Application: Senior Application Systems Engineer

Work Sample 1 - <u>UI Enhancement and Legacy Modernization</u>

- **Objective:** Modernized basic, outdated legacy screens into clean, user-friendly layouts using HTML, CSS, JavaScript, and Bootstrap.
- T**he Challenge:** Modifying presentation elements risked breaking back-end functionalities, server-side data event triggers, and page life-cycle execution.
- **My Strategy:**
  - Separated UI styles from back-end logic by migrating inline elements into modular, standalone CSS sheets.
  - Preserved original DOM structures and element IDs to ensure all JavaScript functionalities and form handlers remained 100% stable.
- **Result & Quality Control:** Prevented code regression during modernization and successfully delivered an enhanced user experience without breaking a single production system function.

Before: Legacy Layout



After: Modernized UI Layout



Work Sample 2: <u>QA Testing & System Functionality Verification.</u>

- **Task:** Conducted functional testing, bug tracking, and user access troubleshooting for the STRIX website (an internal employee skills registry form).
- Risk Managed (The Help Provided):
  - Isolated and resolved complex login and user authentication issues. Ensuring secure employee data registration.
  - Performed rigorous functional testing on core web components to prevent system crashes and layout breaks before updates were pushed.
- **Outcome:** Maintained detailed technical documentation and developed tracking tools to ensure 100% system reliability and a smooth workflow for all internal personnel.

Work Sample 3: <u>Automated Quality Engineering & Code Optimization</u>

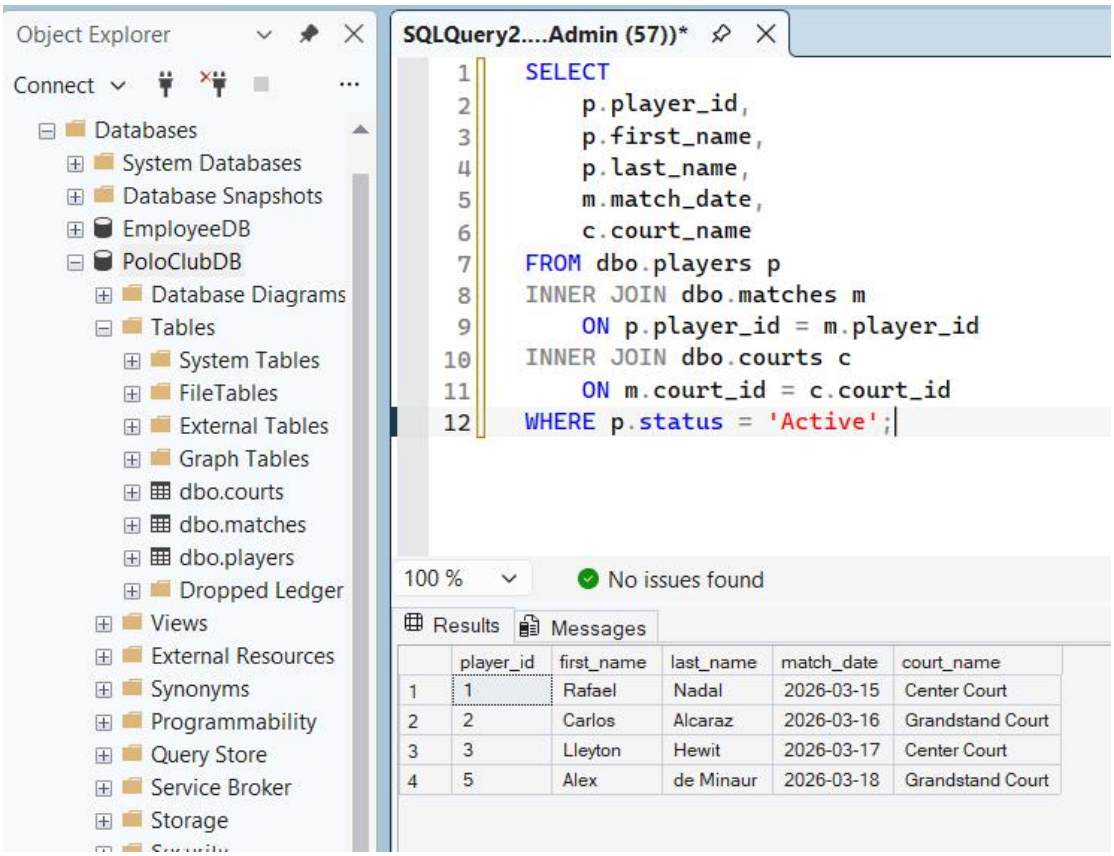
- **Task:** Conducted functional testing and built automated unit test cases for a Java-based development project using Java, JUnit Framework, and Eclipse IDE.
- Risk Managed:

- Perform rigorous code reviews to identify and eliminate redundant logic, ensuring the platform consistently met-high performance standards.
- Isolated and resolved client-side performance bottlenecks, preventing slow system runtime and logical errors before software deployment.

- **Outcome:** Successfully optimized the core codebase efficiency, resulting in a cleaner, faster, and more reliable application framework.

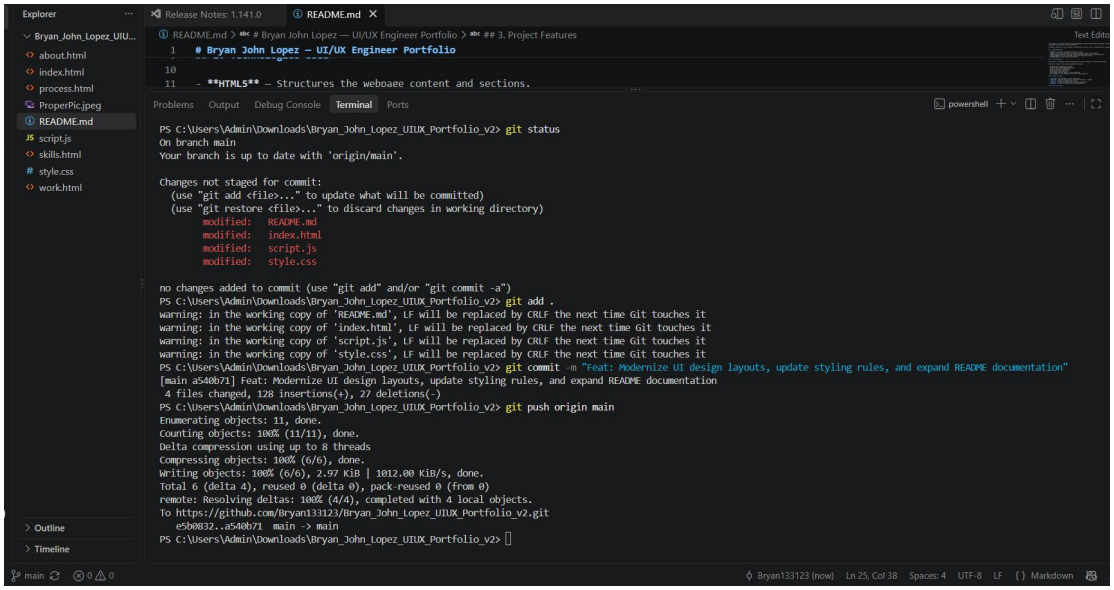
Work Sample 4: <u>Relational Database Design and Multi-Table Data Joining</u>

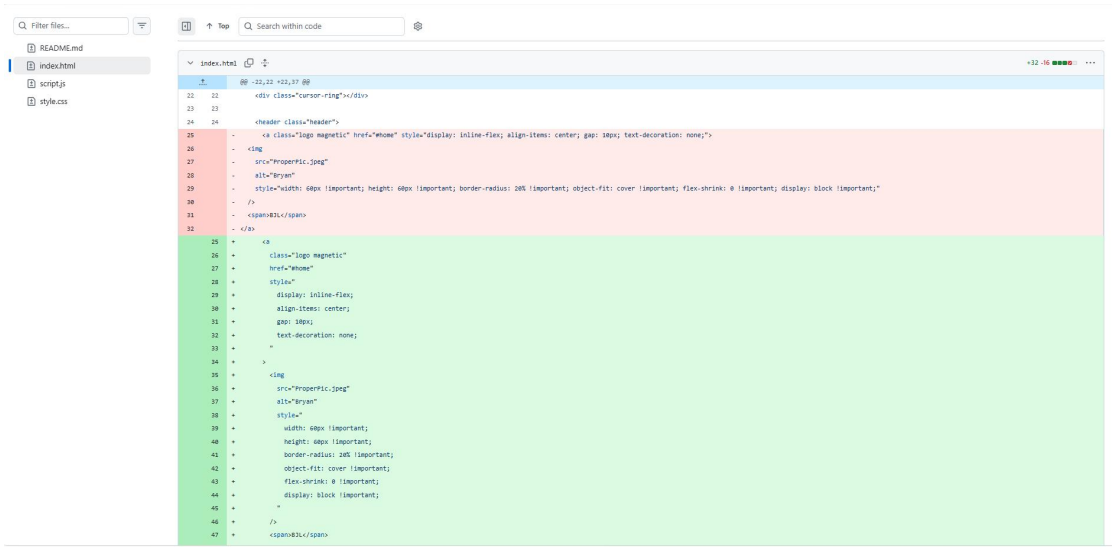
- **Task:** Architected relational database structures and built optimized multi-table validation queries within Microsoft SQL Server (SSMS) to track athletic data feeds.
- **Risk Managed (The strategy implemented):**
  - Prevented system logic errors and execution delays by enforcing strict primary-foreign key relationships across players, matches, and courts entity tables.
  - Designed structured INNER JOIN queries utilizing explicit table aliases and conditional filters (WHERE status = 'Active') to protect against heavy server processing loads and orphaned record reads.
- **Outcome:** Successfully mapped dynamic match schedules directly to live player profiles and court availability data, ensuring seamless real-time application updates.

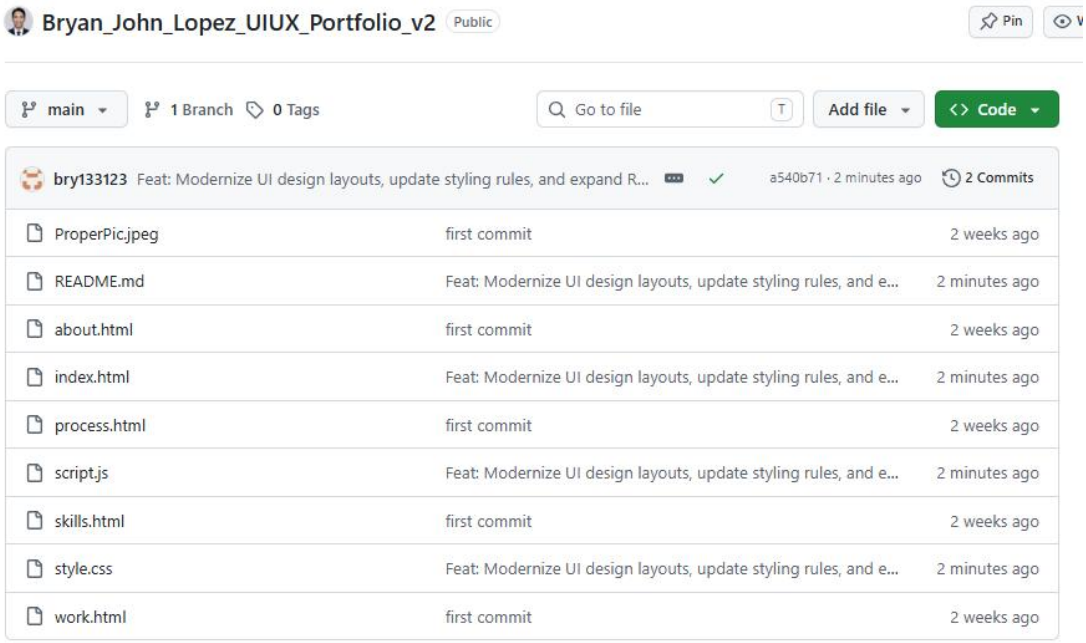


Work Sample 5: <u>Version Control, Code Staging, & DevOps Management</u>

- **Task:** Implemented repository isolation and full code lifecycle staging deployment workflows using Git and Visual Studio Code Terminal (PowerShell).
- **Risk Managed:**
  - Successfully managed repository environment errors (resolving fatal: not a git repository flags) by correctly targeting localized working trees and enforcing core operational tracks.
  - Executed atomic codebase staging using git add . while actively mitigating cross-platform formatting warnings (LF to CRLF normalization) to protect branch stability.
- **Outcome:** Committed and pushed dynamic user interface assets (index.html, style.css, script.js) alongside expanded documentation architectures (README.md) seamlessly into the GitHub remote branch.







README

Bryan John Lopez — UI/UX Engineer Portfolio

1. Project Overview

This project is a personal portfolio website developed to present my professional background, technical skills, and experience in UI/front-end development and application systems engineering.

The website provides a central location for presenting my profile, skills, and selected work through a web-based user interface.

2. Technologies Used